Hafeez Khan

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Education

- **Florida Institute of Technology** — Florida, USA
  Masters & Ph.D. in Computer Science (PhD Advisor: Prof. Siddhartha Bhattacharyya) — *2023 - 2027 (Expected)*
- **Birla Institute of Technology and Science, Pilani** — Dubai, UAE
  Bachelors of Engineering in Computer Science (Thesis Advisor: Prof. Raja Muthalaghu) — *2019 - 2023*

Recent Achievement

1st place Winner, Meta-hosted Competition on Computationally Optimal Gaussian Splatting at ICCV 2025

Developed a lossy compression algorithm leveraging pruning and quantization, reducing 3DGS model sizes by **95.6%**, with rendering performance evaluated on the FlashGS renderer.

Research Experience

Incoming Systems Engineering Intern at Collins Aerospace — Summer 2025
Guide: Janet Liu, Cong Liu — ***RTX, Collins Aerospace, USA***

- Developing an Agentic Vision-Language Model for runtime monitoring of aircraft in X-Plane simulation environment.

Research Assistant at ASSIST Lab (Florida Tech) — Fall 2023 – Present

- **NASA-Funded Research** — Fall 2024 – Present
  Guide: Natasha Neogi, Sarah Lehman, Siddhartha Bhattacharyya, Philip Chan — ***NASA LaRC, USA***
  - Topics: Self-Supervised Learning, Multi-Modal Learning (Text and Vision, Video, Speech), Continual Learning
  - Developed a Multi-Modal Self-Supervised Learning Model that integrates within the generative framework, and outperforms existing state-of-the-art on ImageNet-1k by **2.1%** Top-1 Accuracy.
  - Created a continual learning model for lane detection, reaching optimal accuracy using just **41%** of parameters.
  - Developed automated labeling algorithms to successfully label over **500,000** taxiways and runways images, improving detection accuracy by **7.31%** and reducing per-frame inference latency by **58.4%**.
- **Microsoft-Collaboration Research** — Spring 2025
  Guide: Yash Jain, Vibhav Vineet — ***Microsoft Research, USA***
  - Topics: Image Generation, Multi-modal Large Language Models, Text-to-Image, Prompt Optimization
  - Developed a closed-loop test-time prompt refinement framework using multi-modal LLMs, improving text-to-image alignment by **11.7%** on DALL-E 3, **12.9%** on Flux, and upto **15.25%** on Stable Diffusion 1.5/2.1/3.0 models.
- **DARPA-Funded Research** — Summer 2025
  Guide: Junaid Babar, Isaac Amundson, Siddhartha Bhattacharyya — ***RTX, Collins Aerospace, USA***
  - Topics: Trust and Safety, Formal Verification, Reinforcement Learning, Model Checking, Language Translation
  - Built an automated Soar-to-PRISM translator in ANTLR/Java to enable formal verification of cognitive models.
- **American Heart Association-Funded Research** — Summer 2024
  Guide: Venkat Keshav Chivukula — ***UTHealth Houston, USA***
  - Topics: 3D/4D Reconstruction, Representation Learning, Computational Fluid Dynamics (CFD)
  - Developed an autoencoder for 3D intraventricular velocity reconstruction, surpassing UNet3D by **10.55%** in PSNR.
  - Generated synthetic CFD hemodynamic datasets across **8** LVAD patient geometries using ANSYS Fluent for training 3D models.

Awards

- Awarded for **Outstanding Academic Achievement** at Florida Tech. — 2025
- **Undergraduate Research Award** for outstanding Bachelors Thesis at BITS Pilani Dubai. — 2023
- **Gold Medalist** in IEEE Robotics Competition, 3D Object Detection and Tracking for Road Safety, held in Dubai. — 2023

Skills

Programming Languages: Python, R, Matlab, C/C++, Java, SQL

Proficient Deep Learning and Vision Libraries: PyTorch, TensorFlow, Diffusers, MMDetection, OpenCV, Open3D

Cloud Platforms & Services: Azure (ML Studio, DevOps), AWS (S3, SageMaker), GCP (Google Maps, Cloud Storage).

Publications

∗ equal contribution

Adapt, But Don't Forget: Fine-Tuning and Contrastive Routing for Lane Detection under Distribution Shift [PDF]
M.A. Hafeez Khan, Parth Ganeriwala, Sarah M. Lehman, Siddhartha Bhattacharyya, Amy Alvarez, Natasha Neogi
[Oral Presentation] International Conference on Computer Vision 2COOOL Workshop (**ICCV**), 2025

AssistTaxi-v2: A Scalable Dataset for Taxiway/Runway Scene Understanding in Diverse Conditions [PDF]
Parth Ganeriwala, M.A. Hafeez Khan, Amy Alvarez, Siddhartha Bhattacharyya, Natasha Neogi, Sarah M. Lehman
American Institute of Aeronautics and Astronautics (**AIAA**) SCITECH Forum, 2026

Test-time Prompt Refinement for Text-to-Image Models [PDF]
M.A. Hafeez Khan*, Yash Jain*, Siddhartha Bhattacharyya, Vibhav Vineet
[Invited Talk] International Conference on Computer Vision MARS2 Workshop (**ICCV**), 2025

Few-Shot Classification and Anatomical Localization of Tissues in SPECT Imaging [PDF]
M.A. Hafeez Khan, Samuel M. Boddepalli, Siddhartha Bhattacharyya, Debasis Mitra
[Oral Presentation] Medical Imaging Conference (**MIC**), 2025

LVADNet3D: A Deep Autoencoder for Reconstructing 3D Intraventricular Flow from Sparse Data [PDF]
M.A. Hafeez Khan, Marcello Mattei, Ben Diaz, Ruth White, Siddhartha Bhattacharyya, Venkat Keshav Chivukula
[Oral Presentation] International Conference on Machine Learning and Applications (**ICMLA**), 2025

NORA: A Nephrology-Oriented Representation Learning Approach Towards CKD Classification [PDF]
M.A. Hafeez Khan, T. Bhattacharyya, O. Khan, Noorah Khan, Alina Khan, M.Q. Khan, Sujoy Hajra
[Spotlight Paper] International Conference on Machine Learning and Applications (**ICMLA**), 2025

ALINA: Advanced Line Identification and Notation Algorithm [PDF]
M.A. Hafeez Khan, Parth Ganeriwala, Siddhartha Bhattacharyya, Natasha Neogi, Raja Muthalagu
IEEE / CVF Computer Vision and Pattern Recognition Conference (**CVPR**), 2024

A Hybrid BiLSTM-CNN Approach for Intrusion Detection for IoT Applications [PDF]
M.A. Hafeez Khan, Sapna Sadhwani, Raja Muthalagu, Pranav Pawar, K. Suresh
Scientific Reports, Springer, 2024

AssistTaxi: A Comprehensive Dataset for Taxiway Analysis and Autonomous Operations [PDF]
Parth Ganeriwala, Siddhartha Bhattacharyya, S. Gunther, Brian Kish, M.A. Hafeez Khan, Aknur Dhadoti, Natasha Neogi
International Conference on Machine Learning and Applications (**ICMLA**), 2023

Classification of Microstructure Images of Metals Using Transfer Learning [PDF]
M.A. Hafeez Khan, Hrishikesh Sabnis, J. Angel Arul Jothi, J. Kanishkha, A.D. Prasad
International Conference on Modelling and Development of Intelligent Systems (**MDIS**), 2022

Detection of Cavities from Oral Images using Convolutional Neural Networks [PDF]
M.A. Hafeez Khan, Giri Prasad S., J. Angel Arul Jothi
[Best Paper Award] IEEE International Conference on Electrical, Computer and Energy Technologies (**ICECET**), 2022

Detection of Bicep Form Using Myoware and Machine Learning [PDF]
M.A. Hafeez Khan, Rohan V. Rudraraju, R. Swarnalatha
International Conference on Advances in Data-driven Computing and Intelligent Systems (**ADCIS**), 2022

Teaching Experience

Graduate Teaching Assistant — Florida Tech
Primary Instructor: Prof. Philip Chan (FLTech) — *Aug '23 – May '24*

- Course: CSE2010 Algorithms & Data Structures; My Rating: 4.9/5.